



# Weighted $h^*$ -polynomials of large dilates of smooth polytopes

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Figure: This is joint work with Juliet Whidden and Josephine Yu.

It is also a follow-up to our previous paper:  
<https://arxiv.org/abs/2511.09744>

1. (Weighted) Ehrhart Theory
2. Two Universal Bound Questions
3. Interlude: 2-Dimensional Alcovet Polytopes
4. Results
  - Question 1: Fix  $P$ , vary  $w$
  - Question 2: Fix  $w$ , vary  $P$
  - Future Questions

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# An introduction to polytopes



A (integral) polytope  $P \subseteq \mathbb{R}^d$  is defined as the convex hull of a finite set of points, equivalently, **the closed bounded intersection of finitely many half-spaces**. Equivalently,  $P$  is the solution set to  $Ax \leq \mathbf{b}$  for some (integral) matrix  $A \in \mathbb{Z}^{n \times d}$  and (integral) vector  $\mathbf{b} \in \mathbb{Z}^n$ .

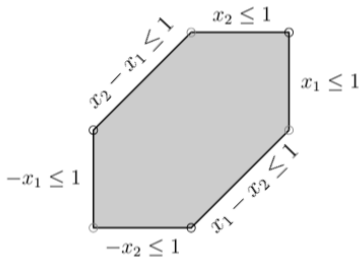


Figure: A example of an polytope. Modified from Joswig and Kulas [2010].

# Ehrhart Theory (1)



"The central problem of Ehrhart Theory is counting the number of **integer points** in **integer dilates** of **integral polytopes**."

**Integral polytopes** are polytopes with all integral vertices.

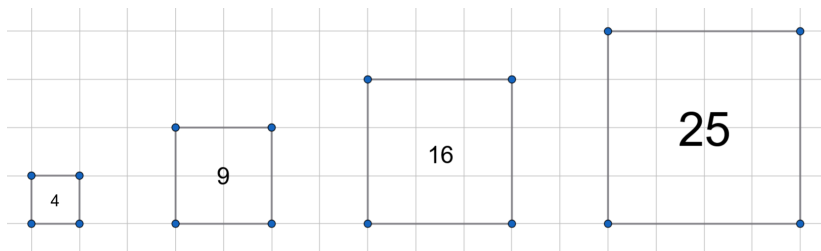


Figure: The first four integer dilates of the square  $P$  with vertices  $(0, 0)$ ,  $(0, 1)$ ,  $(1, 0)$ ,  $(1, 1)$  and their corresponding integer point counts.

**Question for the audience:** In this case, what is  $\#(tP \cap \mathbb{Z}^d)$ ?

## Theorem (Ehrhart Polynomial, Ehrhart [1967])

For any integral  $d$ -dimensional polytope  $P$ ,  $\text{ehr}_P(t) = \#(tP \cap \mathbb{Z}^d)$  is a degree  $d$  polynomial, which we refer to as the **Ehrhart polynomial**.

## Definition (Ehrhart Series)

Moreover, if we encode the numbers  $\text{ehr}_P(0), \text{ehr}_P(1), \text{ehr}_P(2), \dots$  as the coefficients of a power series, we get the **Ehrhart series**

$$\text{Ehr}_P(z) = \sum_{t \geq 0} \text{ehr}_P(t) z^t = \frac{h_d^* z^d + \dots + h_0^*}{(1 - z)^{d+1}}$$

We refer to the numerator  $h_P^*(z) = h_d^* z^d + \dots + h_0^*$  as the  **$h^*$ -polynomial**.

# Ehrhart Theory (2.5)

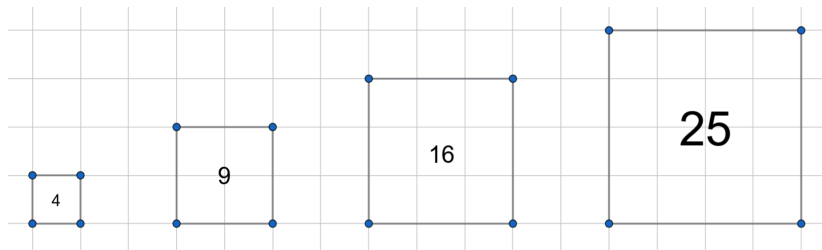


Figure: The Ehrhart polynomial of  $P$  in this case is  $\mathbf{ehr}_P(t) = (t + 1)^2$  and the corresponding Ehrhart series is  $\mathbf{Ehr}_P(z) = 1 + 4z + 9z^2 + 16z^3 + 25z^4 + \dots = (z + 1)/(1 - z)^3$ .

# Weighted Ehrhart Theory (1)



Now what happens if we “weigh” each of our points by a polynomial function?

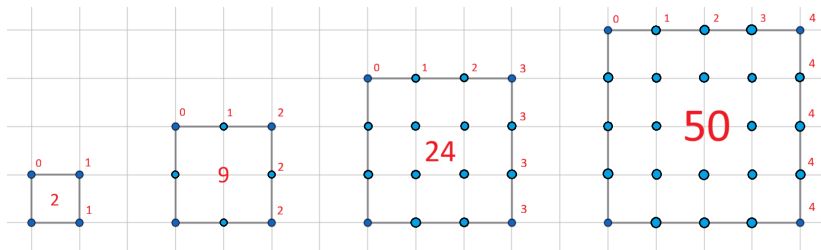


Figure: The first four integer dilates of the square  $P$  with vertices  $(0, 0)$ ,  $(0, 1)$ ,  $(1, 0)$ ,  $(1, 1)$  and their corresponding **weighted** integer point counts according to the function  $w = x_1$ .

# Weighted Ehrhart Theory (2)



## Definition (Weighted Ehrhart Polynomial)

For any integral  $d$ -dimensional polytope  $P$  and homogeneous polynomial  $w$  of degree  $m$ ,  $\text{ehr}_{P,w}(t) = \sum_{p \in tP \cap \mathbb{Z}^d} w(p)$  is a degree  $d + m$  polynomial, called the **weighted Ehrhart polynomial**.

## Definition (Weighted Ehrhart Series)

Moreover, if we encode  $\text{ehr}_{P,w}(0), \text{ehr}_{P,w}(1), \text{ehr}_{P,w}(2), \dots$  as the coefficients of a power series, we get the **weighted Ehrhart series**

$$\text{Ehr}_{P,w}(z) = \sum_{t \geq 0} \text{ehr}_{P,w}(t) z^t = \frac{h_{d+m}^* z^{d+m} + \dots + h_0^*}{(1-z)^{d+m+1}}$$

We refer to the numerator  $h_{P,w}^*(z) = h_{d+m}^* z^{d+m} + \dots + h_0^*$  as the **weighted  $h^*$ -polynomial**.

# Weighted Ehrhart Theory (2.5)



Now what happens if we “weigh” each of our points by a polynomial function?

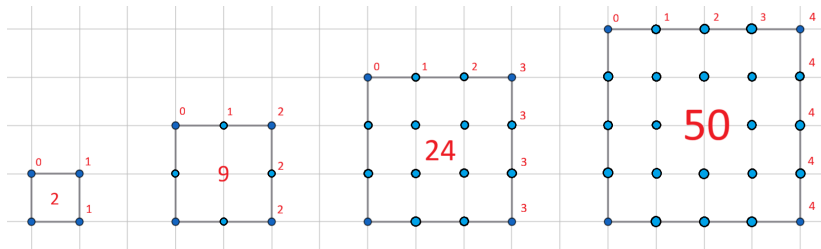


Figure: The weighted Ehrhart polynomial of  $P$  and  $w$  in this case is  $\text{ehr}_{P,w}(t) = (t+1) \sum_{x_1=0}^t x_1 = \frac{1}{2}t(t+1)^2$

and the corresponding weighted Ehrhart series is

$$\text{Ehr}_{P,w}(z) = 2z + 9z^2 + 24z^3 + 50z^4 + \dots = (z^2 + 2z)/(1-z)^4.$$

## Theorem (Stanley [1980])

*For any  $d$ -dimensional integral polytope  $P$ ,  $h_P^*(z)$  has nonnegative integer coefficients.*

However, this is not true in the weighted case! Not even for specialized cases of polytopes and weights!

## Original Motivating Question

Which combinations of integral polytopes  $P$  and homogeneous weight functions  $w$  have weighted  $h^*$ -polynomials  $h_{P,w}^*(z)$  with nonnegative coefficients?

**Today's question: What happens to weighted  $h^*$ -polynomials of dilates of  $P$ :  $2P, 3P, 4P, 5P, \dots$ ?**

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# Root Convergence (1)



Say we dilate our polytope  $P$  by an arbitrary integer factor  $r$ . We will refer to it as  $rP$ .

Theorem (Brenti and Welker [2009])

*The roots of  $h_{rP}^*(z)$  converge towards the roots of the Eulerian polynomial  $A_d(z)$  as  $r \rightarrow \infty$ .*

This holds in the weighted case too!

Theorem (H., Whidden, Yu (2025))

*The roots of  $h_{rP,w}^*(z)$  converge towards the roots of the Eulerian polynomial  $A_{d+m}(z)$  as  $r \rightarrow \infty$ . Moreover, the coefficients of  $h_{rP,w}^*(z)$  eventually have the same sign for large enough  $r$ .*

## Definition ( $h^{<r>}(z)$ )

Let  $f(t) = \sum_{i=0}^k c_i t^i$  be a degree  $k$  polynomial. We can rewrite

$$\sum_{t \geq 0} f(t) z^t = \frac{h(z)}{(1-z)^{k+1}}$$

We define the closely related polynomial  $h^{<r>}(z)$ :

$$\sum_{t \geq 0} f(rt) z^t = \frac{h^{<r>}(z)}{(1-z)^{k+1}}$$

If  $f(t)$  represents the weighted Ehrhart polynomial of  $P$  and  $w$ , then  $h^{<r>}(z)$  is the weighted  $h^*$ -polynomial of  $rP$  and  $w$ .

## Theorem (H., Whidden, Yu (2025))

***The roots of  $h_{rP,w}^*(z)$  converge towards the roots of the Eulerian polynomial  $A_{d+m}(z)$  as  $r \rightarrow \infty$ . Moreover, the coefficients of  $h_{rP,w}^*(z)$  eventually have the same sign for large enough  $r$ .***

## Proof.

Let  $f(t) = \sum_{i=0}^k c_i t^i$  be a degree  $k$  polynomial. We can rewrite

$$h^{<r>}(z) = c_k r^k A_k(z) + c_{k-1} r^{k-1} A_{k-1}(z) + \dots + c_0 (1-z)^k$$

As  $r \rightarrow \infty$ , the roots of  $h^{<r>}(z)$  converge towards the roots of  $A_k(z)$ . Setting  $f(t)$  to our weighted Ehrhart polynomial completes the proof. □

## Theorem (H., Whidden, Yu (2025))

*The roots of  $h_{rP,w}^*(z)$  converge towards the roots of the Eulerian polynomial  $A_{d+m}(z)$  as  $r \rightarrow \infty$ . **Moreover, the coefficients of  $h_{rP,w}^*(z)$  eventually have the same sign for large enough  $r$ .***

## Proof.

It follows that  $h_{rP,w}^*(z)$  is **real rooted with simple negative real roots** for sufficiently high  $r$ , implying the coefficients of  $h_{rP,w}^*(z)$  are **all the same sign**. Moreover, the absolute values of the coefficients of  $h_{rP,w}^*(z)$  are **log-concave** and thus **unimodal**.



Let's return back to unweighted Ehrhart theory for a second.

## Motivating Question

For any  $d$ -dimensional integral polytope  $P$ , is there a universal bound  $R$  such that  $h_{rP}^*(z)$  is simple negative real rooted for all  $r \geq R$ ?

Theorem (Conjectured by Beck and Stapledon [2010], proven by Jochemko [2018])

*$h_{rP}^*(z)$  is simple negative real rooted for all  $r \geq d$ .*

Proof involves carefully creating a family of interlacing polynomials.

## Motivating Question (Weighted Version)

For any  $d$ -dimensional integral polytope  $P$  and homogeneous weight  $w$  of degree  $m$ , is there a universal bound  $R$  such that  $h_{rP,w}^*(z)$  is simple negative real rooted for all  $r \geq R$ ?

This question is too specific, let's allow  $P$  and  $w$  to vary.

## Question 1: Fix $P$ , vary $w$

Fix a  $d$ -dimensional integral polytope  $P$  and degree  $m$ . For all degree  $m$  homogeneous weights  $w$ , is there a universal bound  $N_{P,m}$  such that  $h_{rP,w}^*(z)$  is simple negative real rooted for all  $r \geq N_{P,m}$ ?

## Question 2: Fix $w$ , vary $P$

Fix a homogeneous weight  $w$  and dimension  $d$ . For all integral  $d$ -dimensional polytopes  $P$ , is there a universal bound  $N_{w,d}$  such that  $h_{rP,w}^*(z)$  is simple negative real rooted for all  $r \geq N_{w,d}$ ?

# Counterevidence?



We

found a 2-dimensional polytope  $P$  and degree 1 weight  $w = x_1$  that required 63 dilates before all coefficients were nonnegative:

- $h_{P,w}^*(z) = -10z^3 + 11z$
- $\vdots$
- $h_{62P,w}^*(z) = -620z^3 + 158844z^2 + 80104z$
- $h_{63P,w}^*(z) = 21z^3 + 166656z^2 + 83370z$

Source: <https://sites.gatech.edu/weighteddehrhart>

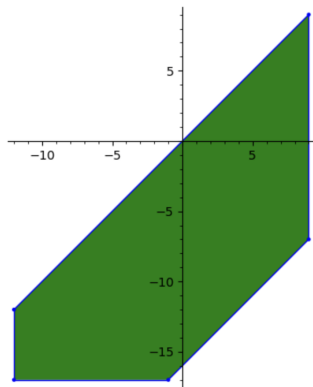


Figure: Our polytope  $P$  described on the left.

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# 2-Dimensional Alcoved Polytopes



2-Dimensional Alcoved Polytopes are defined as the intersection of the half-spaces

$$x - y \leq \mathbf{b}_{12}$$

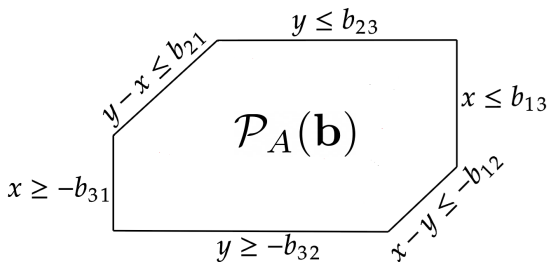
$$x \leq \mathbf{b}_{13}$$

$$y - x \leq \mathbf{b}_{21}$$

$$y \leq \mathbf{b}_{23}$$

$$-x \leq \mathbf{b}_{31}$$

$$-y \leq \mathbf{b}_{32}$$



such that the triangle inequalities  $\mathbf{b}_{ij} + \mathbf{b}_{jk} \geq \mathbf{b}_{ik}$  are satisfied. We will denote these alcoved polytopes as  $\mathcal{P}_A(\mathbf{b})$  for this talk.

## Definition (Smooth Polytopes)

A  $d$ -dimensional integral polytope  $P \subseteq \mathbb{R}^d$  is called **smooth** if each vertex cone of  $P$  is generated by a basis of  $\mathbb{Z}^d$ .

## Theorem (H., Whidden, Yu (2025))

*There exists an algorithm that when given a fixed smooth polytope  $P_A(\mathbf{b}^0)$  and homogeneous weight function  $w$ , explicitly computes the parametric weighted Ehrhart and  $h^*$ -polynomials for smooth polytopes  $P_A(\mathbf{b})$  defined by the same facets as  $P_A(\mathbf{b}^0)$  in terms of the “right-hand side variables”  $\mathbf{b}_i$  of  $P$ .*

As a result, we can compute the weighted Ehrhart polynomials and  $h^*$ -polynomials of 2-dimensional alcoved polytopes!

# Parametric Formula Example



Let  $P_A(\mathbf{b})$  be the 2-dimensional alcoved polytope and  $w = x_1$ .

$$\begin{aligned}
 h_{P_A(\mathbf{b}), w}^*(z) = & -\frac{1}{6} \mathbf{b}_{12}^3 z^3 + \frac{1}{2} \mathbf{b}_{12} \mathbf{b}_{13}^2 z^3 - \frac{1}{3} \mathbf{b}_{13}^3 z^3 + \frac{1}{6} \mathbf{b}_{21}^3 z^3 + \frac{1}{2} \mathbf{b}_{13}^2 \mathbf{b}_{23} z^3 - \frac{1}{2} \mathbf{b}_{21}^2 \mathbf{b}_{23} z^3 \\
 & + \frac{1}{2} \mathbf{b}_{21} \mathbf{b}_{23}^2 z^3 - \frac{1}{6} \mathbf{b}_{23}^3 z^3 - \frac{1}{2} \mathbf{b}_{21} \mathbf{b}_{31}^2 z^3 + \frac{1}{3} \mathbf{b}_{31}^3 z^3 + \frac{1}{2} \mathbf{b}_{12}^2 \mathbf{b}_{32} z^3 - \frac{1}{2} \mathbf{b}_{31}^2 \mathbf{b}_{32} z^3 \\
 & - \frac{1}{2} \mathbf{b}_{12} \mathbf{b}_{32}^2 z^3 + \frac{1}{6} \mathbf{b}_{32}^3 z^3 - \frac{2}{3} \mathbf{b}_{12}^3 z^2 + 2 \mathbf{b}_{12} \mathbf{b}_{13}^2 z^2 - \frac{4}{3} \mathbf{b}_{13}^3 z^2 + \frac{2}{3} \mathbf{b}_{21}^3 z^2 + 2 \mathbf{b}_{13}^2 \mathbf{b}_{23} z^2 \\
 & - 2 \mathbf{b}_{21}^2 \mathbf{b}_{23} z^2 + 2 \mathbf{b}_{21} \mathbf{b}_{23}^2 z^2 - \frac{2}{3} \mathbf{b}_{23}^3 z^2 - 2 \mathbf{b}_{21} \mathbf{b}_{31}^2 z^2 + \frac{4}{3} \mathbf{b}_{31}^3 z^2 + 2 \mathbf{b}_{12}^2 \mathbf{b}_{32} z^2 - 2 \mathbf{b}_{31}^2 \mathbf{b}_{32} z^2 \\
 & - 2 \mathbf{b}_{12} \mathbf{b}_{32}^2 z^2 + \frac{2}{3} \mathbf{b}_{32}^3 z^2 - \frac{1}{2} \mathbf{b}_{12} \mathbf{b}_{13} z^3 - \frac{1}{2} \mathbf{b}_{13} \mathbf{b}_{23} z^3 + \frac{1}{2} \mathbf{b}_{21} \mathbf{b}_{31} z^3 + \frac{1}{2} \mathbf{b}_{31} \mathbf{b}_{32} z^3 \\
 & - \frac{1}{6} \mathbf{b}_{12}^3 z + \frac{1}{2} \mathbf{b}_{12} \mathbf{b}_{13}^2 z - \frac{1}{3} \mathbf{b}_{13}^3 z + \frac{1}{6} \mathbf{b}_{21}^3 z + \frac{1}{2} \mathbf{b}_{13}^2 \mathbf{b}_{23} z - \frac{1}{2} \mathbf{b}_{21}^2 \mathbf{b}_{23} z + \frac{1}{2} \mathbf{b}_{21} \mathbf{b}_{23}^2 z \\
 & - \frac{1}{6} \mathbf{b}_{23}^3 z - \frac{1}{2} \mathbf{b}_{21} \mathbf{b}_{31}^2 z + \frac{1}{3} \mathbf{b}_{31}^3 z + \frac{1}{2} \mathbf{b}_{12}^2 \mathbf{b}_{32} z - \frac{1}{2} \mathbf{b}_{31}^2 \mathbf{b}_{32} z - \frac{1}{2} \mathbf{b}_{12} \mathbf{b}_{32}^2 z + \frac{1}{6} \mathbf{b}_{32}^3 z \\
 & + \frac{1}{6} \mathbf{b}_{12} z^3 + \frac{1}{3} \mathbf{b}_{13} z^3 - \frac{1}{6} \mathbf{b}_{21} z^3 + \frac{1}{6} \mathbf{b}_{23} z^3 - \frac{1}{3} \mathbf{b}_{31} z^3 - \frac{1}{6} \mathbf{b}_{32} z^3 + \frac{1}{2} \mathbf{b}_{12} \mathbf{b}_{13} z \\
 & + \frac{1}{2} \mathbf{b}_{13} \mathbf{b}_{23} z - \frac{1}{2} \mathbf{b}_{21} \mathbf{b}_{31} z - \frac{1}{2} \mathbf{b}_{31} \mathbf{b}_{32} z - \frac{1}{3} \mathbf{b}_{12} z^2 - \frac{2}{3} \mathbf{b}_{13} z^2 + \frac{1}{3} \mathbf{b}_{21} z^2 - \frac{1}{3} \mathbf{b}_{23} z^2 + \dots
 \end{aligned}$$

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## Question 1: Fix $P$ , vary $w$

Fix a  $d$ -dimensional integral polytope  $P$  and degree  $m$ . For all degree  $m$  homogeneous weights  $w$ , is there a universal bound  $N_{P,m}$  such that  $h_{rP,w}^*(z)$  is simple negative real rooted for all  $r \geq N_{P,m}$ ?

## Definition ( $L_{P,m}$ )

Define the map  $w \rightarrow h_{P,w}^*$  from the set of all degree  $m$  homogeneous weights  $w$  to their respective weighted  $h^*$ -polynomials. Denote the range of this map as  $L_{P,m}$ .

## Lemma (H., Whidden, and Yu (2026+))

*If rank  $L_{P,m} \geq 2$ , then there exists some homogeneous degree  $m$  weight function  $w$  such that  $h_{P,w}^*$  has mixed sign coefficients and is thus not simple negative rooted.*

## Proof.

We view the polynomials in  $L_{P,m}$  as vectors in  $\mathbb{R}^{d+m}$  by mapping the coefficient of  $z^i$  in  $h_{P,w}^*$  to the  $i^{\text{th}}$  entry of its corresponding vector. If rank  $L_{P,m} \geq 2$ , then it must contain two polynomials  $h_{P,w_1}^*(z) \rightarrow \vec{a}$  and  $h_{P,w_2}^*(z) \rightarrow \vec{b}$  whose coefficient vectors are linearly independent with nonnegative entries. Since  $\vec{a}$  and  $\vec{b}$  are linearly independent, we can find some linear combination  $\vec{a} - c\vec{b}$  with mixed sign coefficients. □

# $2 \times 2$ Minors Observation (1)



## Question 1: Fix $P$ , vary $w$

Fix a  $d$ -dimensional integral polytope  $P$  and degree  $m$ . For all degree  $m$  homogeneous weights  $w$ , is there a universal bound  $N_{P,m}$  such that  $h_{rP,w}^*(z)$  is simple negative real rooted for all  $r \geq N_{P,m}$ ?

**Observation: A universal bound cannot exist if rank  $L_{P,m} \geq 2$ .**

Okay, so when does rank  $L_{P,m} = 1$ ? We can take the  $2 \times 2$  minors of the matrix

$$\begin{bmatrix} \dots & h_{P,x_1^m}^*(z) & \dots \\ \dots & \vdots & \dots \\ \dots & h_{P,x_d^m}^*(z) & \dots \end{bmatrix}$$

and see when they vanish!

# 2 × 2 Minors Observation (2)



## Lemma (H., Whidden, and Yu (2026+))

Let  $P_A(\mathbf{b})$  be the 2-dimensional alcoved polytope and  $m = 1$ . We obtain a characterization of when  $\text{rank } L_{P_A(\mathbf{b}),m} = 1$ :

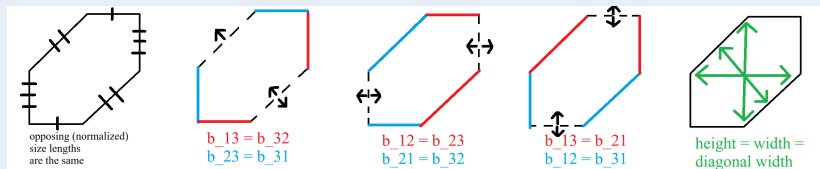
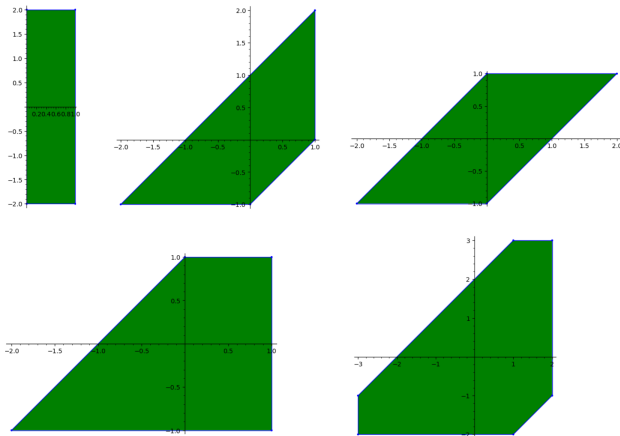


Figure: The five different cases where  $\text{rank } L_{P_A(\mathbf{b}),m} = 1$ .

# $2 \times 2$ Minors Observation (3)



For  $m = 2$ , there is no neat classification...



...and for  $m \geq 3$ , it seems we're out of luck...

For  $m = 2$ , there is no neat classification...



## Question 2: Fix $w$ , vary $P$

Fix a homogeneous weight  $w$  and dimension  $d$ . For all integral  $d$ -dimensional polytopes  $P$ , is there a universal bound  $N_{w,d}$  such that  $h_{rP,w}^*(z)$  is simple negative real rooted for all  $r \geq N_{w,d}$ ?

**Observation: Simple negative real rooted implies nonnegativity.**

For this presentation, we will focus on  $w = x_1$ .

Using our parametric formula, we found a bunch of counterexamples...

...63 dilates, 87 dilates, 120 dilates, 250 dilates, 20,000 dilates...

...and eventually found a family of polytopes breaking the universal bound condition!

# Counterexample $w = x_1$ (1)



## Theorem (H., Whidden, and Yu (2026+))

*There does not exist a universal nonnegativity bound for 2-dimensional polytopes with respect to the weight  $w = x_1$ .*

## Proof.

We found a family of polytopes  $P^{(k)}$  such that

$$\begin{aligned} h_{rP^{(k)},w}^*(z) = & \left( \frac{1}{3}r^3 - \left(k + \frac{1}{2}\right)r^2 + \frac{1}{6}r \right) z^3 \\ & + \left( \frac{4}{3}r^3 - \frac{1}{3}r \right) z^2 + \left( \frac{1}{3}r^3 + \left(k + \frac{1}{2}\right)r^2 + \frac{1}{6}r \right) z \end{aligned}$$

Observe that the leading  $z^3$  coefficient can be made to be negative for any dilate  $r$  by adjusting  $k$ .



# Counterexample $w = x_1$ (1.5)



## Theorem (H., Whidden, and Yu (2026+))

*There does not exist a universal nonnegativity bound for 2-dimensional polytopes with respect to the weight  $w = x_1$ .*

Here are some examples of the parameterization we found:

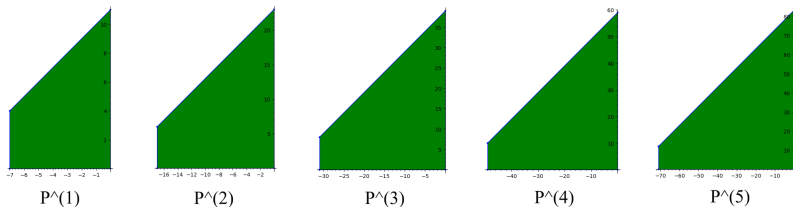


Figure: Examples of our parameterized polytopes  $P^{(k)}$ .

# Counterexample $w = x_1$ (2)



Corollary (H., Whidden, and Yu (2026+))

*There does not exist a universal nonnegativity bound for  $d$ -dimensional polytopes with weight  $w = x_1$  for  $d \geq 3$ .*

Proof.

Take our family of polytopes  $P^{(k)}$  constructed earlier and consider the Cartesian products  $Q^{(k)} = P^{(k)} \times [0, 1]^{d-2}$ . Then the leading coefficient  $z^{d+1}$  of  $h_{Q^{(k)}, w}^*(z)$  is

$$\left(\frac{1}{3}r^3 - \left(k + \frac{1}{2}\right)r^2 + \frac{1}{6}r\right)(r-1)^{d-2}$$

Similarly to earlier, we can set  $k$  arbitrarily large to ensure that the above is negative for any arbitrary dilate of  $r$ . We also note that the sum of the coefficients is  $h_{Q^{(k)}, w}^*(1)$  is nonnegative.  $\square$

- Constructing counterexamples for other families of polytopes and monomial weights, ex:  $w = x_1^2$  and  $w = x_1x_2$ .
  - We conjecture that a universal bound exists  $w$  is a product of even powered linear forms.
- Considering when  $w$  is non-monomial, ex:  $w = ax_1 + bx_2$  or the Motzkin polynomial.
- Can we find  $h_{P,w}^*$  such that it has nonnegative coefficients but is NOT unimodal, log-concave, or simple negative real rooted?
  - For our  $P^{(k)}$  example, it turns out that as soon as  $h_{rP^{(k)},w}^*$  has nonnegative coefficients, it is simple negative real rooted.
  - If  $P$  is 1-dimensional, then all  $h_{P,w}^*$  have nonnegative coefficients (H., Whidden, and Yu 2026+). We conjecture they are all simple negative real rooted as well.

- We investigate the behavior of **weighted  $h^*$ -polynomials**  $h_{rP,w}^*$  of the  $r^{\text{th}}$  dilate of a polytope  $rP$  as  $r \rightarrow \infty$ .
- Our main question was: For any  $d$ -dimensional integral polytope  $P$  and homogeneous weight  $w$  of degree  $m$ , is there a **universal bound**  $R$  such that  $h_{rP,w}^*(z)$  is simple negative real rooted for all  $r \geq R$ ?
  - If we fix  $P$  and let  $w$  vary, then we require  $\text{rank } L_{P,m} = 1$ . However, this is hard to quantify.
  - If we fix  $w$  and let  $P$  vary, then we were able to come up with a counterexample for all dimensions  $d \geq 2$  when  $w = x_1$ .
- Future work includes investigating other monomial (and non-monomial) weights and identifying when  $h_{rP,w}^*$  is simple negative real rooted.

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Thank you!



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